Project Report: Customer Segmentation using Unsupervised Machine Learning

Project Title: Customer Segmentation using K-Means, DBSCAN, and Hierarchical Clustering **Category:** Unsupervised Machine Learning
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1. Abstract

Businesses often have rich customer data but no easy way to translate it into strategy. This project applies three unsupervised clustering algorithms — **K-Means**, **DBSCAN**, and **Agglomerative (Hierarchical) Clustering** — to a mall customer dataset containing age, annual income, and spending score, in order to discover natural customer segments. The optimal number of clusters is determined objectively using the **Elbow Method** and **Silhouette Score**. The three algorithms are then compared using internal validation metrics, and the best-performing model (K-Means) is used to generate concrete, plain-language business recommendations for each discovered segment.

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2. Introduction

2.1 Problem Statement

A mall/retail business wants to understand its customer base well enough to run targeted marketing campaigns instead of one-size-fits-all promotions. However, the business has no pre-labeled "customer type" — it only has raw demographic and behavioural data. This is a classic **unsupervised learning** problem: we must discover the structure in the data ourselves.

2.2 Objectives

1. Clean and explore a real-world-style customer dataset.
2. Determine the statistically optimal number of customer segments.
3. Train and rigorously compare three different clustering algorithms.
4. Select the best model using quantitative evaluation metrics.
5. Convert the resulting clusters into actionable business insights.

2.3 Why Unsupervised Learning?

Supervised learning requires labeled outcomes (e.g., "this customer churned" / "did not churn"). Here, there is no ground-truth label for "customer segment" — segments are, by definition, something we

want the data to reveal. Clustering algorithms group customers by similarity in feature space without needing any labels, which is exactly the right tool for this business problem.

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3. Dataset Description

The project uses a dataset structured identically to the widely-used **Mall Customer Segmentation Dataset** available on Kaggle (vjchoudhary7/customer-segmentation-tutorial-in-python).

Column	Type	Description
CustomerID	Integer	Unique identifier per customer
Gender	Categorical	Male / Female
Age	Integer	Customer's age in years
Annual Income (k$)	Integer	Annual income in thousands of USD
Spending Score (1-100)	Integer	Score assigned by the mall based on customer behaviour and spending

Size: 200 customer records (203 before duplicate removal, to simulate real-world data quality issues).

The bundled `data/mall_customers.csv` is a synthetically generated dataset engineered to match the real dataset's statistical structure (five natural income/spending archetypes, similar age and income ranges) plus a handful of injected missing values and duplicate rows, so that the preprocessing pipeline in this project has genuine problems to solve. Swapping in the original Kaggle CSV (same column names) reproduces the same pipeline on real data.

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4. Methodology

4.1 Data Preprocessing

1. **Missing values:** 5 missing values were injected into `Annual Income (k$)`. These were imputed using the **column median**, which is robust to outliers compared to the mean.
2. **Duplicates:** 3 duplicate rows were injected and removed using `drop_duplicates` (ignoring the `CustomerID` column, since IDs are always unique even for duplicate records).
3. **Encoding:** `Gender` was label-encoded (`Female=0`, `Male=1`) for completeness, though it was ultimately excluded from the clustering feature set (see §4.2).
4. **Scaling:** `Age`, `Annual Income (k$)`, and `Spending Score (1-100)` were standardized using `StandardScaler` (zero mean, unit variance). This is essential because all three clustering algorithms used here are **distance-based**, and Income (range ~15–100) would otherwise dominate the distance calculation over Spending Score or Age.

4.2 Feature Selection

Clustering was performed on `Age`, `Annual Income (k$)`, and `Spending Score (1-100)`. Gender was explored during EDA but excluded from the clustering features themselves, since the correlation analysis showed no meaningful relationship between gender and spending behaviour — including it would have added noise without improving segment quality.

4.3 Exploratory Data Analysis

- **Distribution plots** showed Age, Income, and Spending Score are reasonably spread with no extreme skew after cleaning.
- **Correlation heatmap** showed near-zero linear correlation between Annual Income and Spending Score — confirming that income alone cannot predict spending behaviour, which is precisely why a 2D/3D clustering approach adds value over simple income-based segmentation.
- **Pairplots** visually previewed the natural groupings later confirmed by clustering.

4.4 Determining the Optimal Number of Clusters

Two complementary methods were used across `k = 2` to `10`:

- **Elbow Method:** plots K-Means inertia (within-cluster sum of squared distances) against
  k. Inertia always decreases as k increases, so we look for the point of diminishing returns — the "elbow."
- **Silhouette Score:** measures how similar a point is to its own cluster versus other clusters (range: -1 to 1). This gives an objective, single-number way to compare different values of k.

Both methods pointed to **k = 5**: the elbow visibly bends around k=4–5, and the silhouette score peaks at k=5 before declining for higher k.

4.5 Model Training

Three clustering algorithms were trained on the same scaled feature matrix:

1. **K-Means** (`n_clusters=5`, `init="k-means++"`, `n_init=10`, `random_state=42`) — partitions data into k spherical clusters by minimizing within-cluster variance.
2. **DBSCAN** (`eps=0.6`, `min_samples=5`) — a density-based algorithm that discovers the number of clusters automatically and can flag outliers as noise (label `-1`).
3. **Agglomerative (Hierarchical) Clustering** (`n_clusters=5`, `linkage="ward"`) — builds a tree of nested clusters bottom-up, merging the pair that increases within-cluster variance the least at each step. Ward linkage was chosen because it optimizes the same objective as K-Means, making the two directly comparable.

4.6 Evaluation Metrics

- **Silhouette Score** (higher is better): how well-separated and internally cohesive clusters are.
- **Davies-Bouldin Index** (lower is better): ratio of within-cluster

scatter to between-cluster separation.

For DBSCAN, noise points were excluded from metric computation, since they are not assigned to any cluster.

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5. Results

Algorithm	Clusters Found	Silhouette Score ↑	Davies-Bouldin Index ↓
K-Means	5	**0.456**	0.827
DBSCAN	3 (+ noise points)	0.432	**0.767**
Hierarchical	5	0.440	0.863

Observations: - K-Means produced the highest silhouette score, indicating the most internally cohesive, well-separated clusters among the three algorithms on this dataset. - DBSCAN achieved a slightly better Davies-Bouldin index but collapsed into only 3 dense clusters (plus a small number of noise points) — useful for outlier detection, but less useful for a business that wants every customer assigned to an actionable segment. - Hierarchical Clustering produced results very close to K-Means, as expected given the shared Ward-linkage / variance-minimizing objective — and its dendrogram is a valuable visualization tool for presenting the “family tree” of customer similarity to non-technical stakeholders.

Model selected: K-Means (k=5) — best quantitative performance, full coverage of all customers, and directly interpretable centroids.

5.1 Cluster Profiles (K-Means, k=5)

Cluster	Count	Avg. Age	Avg. Income ($k)	Avg. Spending Score	Segment Name
0	~48	~38.5	~51.1	~50.0	Standard / Average Customers
1	~35	~32.6	~75.7	~69.3	Premium / High-Value Customers
2	~37	~48.4	~80.6	~22.0	High Income, Cautious Spenders
3	~40	~26.2	~28.3	~79.4	Budget-Conscious Big Spenders
4	~40	~59.2	~26.8	~21.2	Low Engagement Customers

(Values shown are from the reference run on the bundled synthetic dataset; running the pipeline will reproduce these deterministically due to the fixed random seed, and will differ slightly if the original Kaggle CSV is substituted.)

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6. Business Insights & Recommendations

1. **Premium / High-Value Customers** (moderate-high income, high spending) — the mall's core revenue drivers today. *Recommendation:* loyalty tiers, early access to new collections, premium-brand cross-promotions.
2. **High Income, Cautious Spenders** (high income, low spending) — represent the single largest untapped revenue opportunity. *Recommendation:* trust-building, highly personalized offers (not generic discounts) to convert latent purchasing power into actual spend.
3. **Budget-Conscious Big Spenders** (lower income, high spending, younger skew) — highly engaged despite budget constraints. *Recommendation:* installment/EMI options, bundled value packs, referral incentives — this segment is receptive to volume-based promotions.
4. **Low Engagement Customers** (low income, low spending, older skew) — least active segment. *Recommendation:* simple, clear discount campaigns and win-back offers; avoid over-investing premium marketing budget here.
5. **Standard / Average Customers** — the stable baseline. *Recommendation:* maintain satisfaction through routine seasonal promotions without heavy targeted spend.

This segmentation allows a marketing team to allocate budget where it will generate the highest return, rather than treating all 200+ customers identically.

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7. Conclusion

This project delivered a complete, reproducible unsupervised machine learning pipeline for customer segmentation — from raw, imperfect data through cleaning, exploratory analysis, objective model-selection (Elbow + Silhouette), multi-algorithm comparison, and finally translation into business-usable insights. K-Means with k=5 was selected as the final model based on its superior silhouette score and full customer coverage, while DBSCAN and Hierarchical Clustering served as valuable comparison points — DBSCAN for outlier detection, Hierarchical for its interpretable dendrogram.

8. Future Work

- Incorporate additional behavioural features (purchase frequency, recency, basket size) if available, moving toward full **RFM (Recency, Frequency, Monetary) segmentation**.
- Use **PCA** to visualize clusters when more than 3 features are used.
- Deploy the trained model as a lightweight web dashboard for live customer scoring.
- Monitor **segment migration** over time to measure whether

marketing actions are successfully moving customers toward higher-value segments.
- Experiment with **Gaussian Mixture Models** for probabilistic (soft) cluster membership.

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9. References

1. Mall Customer Segmentation Dataset — Kaggle: https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python
2. Pedregosa, F. et al. (2011). *Scikit-learn: Machine Learning in Python.* Journal of Machine Learning Research, 12, 2825–2830.
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